Final Project Design Phase

Computer Science - Week 11

Oct 3, 2026 - Version 0.0.4

For complete instructions and project ideas, refer to the assignment in Canvas. Specifically, review the final project requirements before you continue below. If you have not reviewed the project requirements in Canvas (you might have to go to the modules view and check the current or past week's content), you will not be able to properly complete the parts below. In addition, it may be helpful to review the information on this Drafter documentation page.

0) Project Planning and Preparation

Successful programmers spend time planning before they begin coding. *After* reviewing the project requirements, answer the following questions to help guide your design and development process.

A. What concepts from class or previous assignments might help you complete this project? Which programming concepts (loops, conditionals, lists, classes, etc.) do you expect to use? What programming skills do you want to improve while completing this project?

I will probably reference the drafter assignments the most, especially how to transfer data between pages and update state values. I expect to use dataclass State, Workout, list of Workout, and for loops to go through the list. I also plan on testing my code pretty thoroughly as I go and I hope to get better at that.

B. Which parts of the project do you expect to be the easiest for you? The hardest? Why?

I expect the easiest parts would be the calculation functions where I would convert time to seconds and finding the 500m split. I expect the hardest part to be connecting the routes so data is correctly held in the State because there are going to be a lot of changes in data

C. Even though you have not yet designed the final project, estimate which tasks will likely require the most time and effort? Which features or components should be implemented first?

The task that will probably take me the most time and effort will be writing unit tests for every route and getting data to go correctly through it. I think implementing the core pieces of the program would be the best way to start, with the State and Workout dataclasses so I know what data I'm collecting. The helper functions for calculating splits should also be done early.

D. What distractions might interfere with your work and how will you minimize them? What resources will you need available while working? What questions, if any, do you still have about the assignment requirements?

My biggest distraction is probably going to be my assignments for other classes. I plan on scheduling blocks of time for project work so it doesn't get pushed aside. While working I'll probably need past assignments or llm tools if I get stuck.

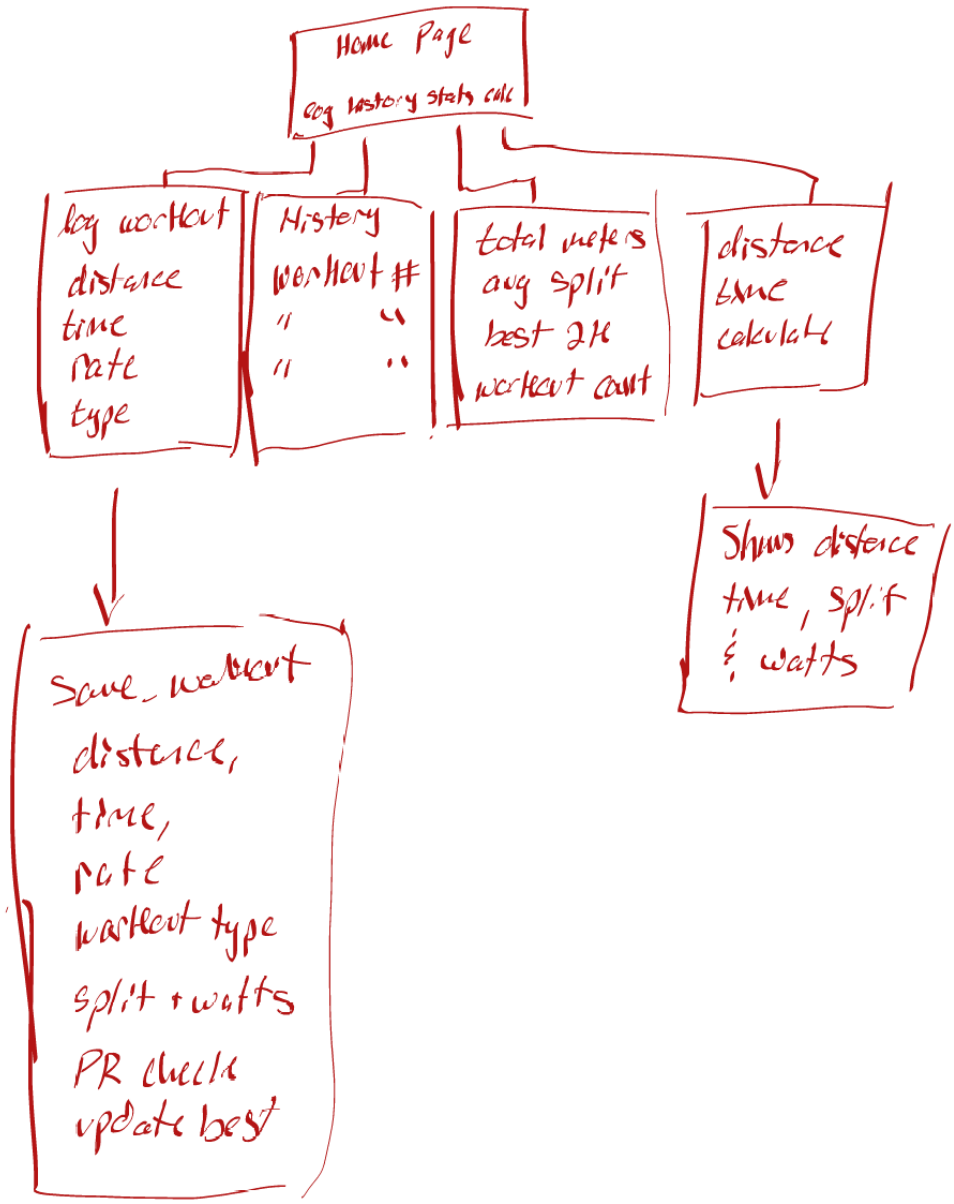
1) Site Diagram

Create a mock-up of your planned website. We suggest you make the mockup on paper, although you can also use software like PowerPoint or Slides to make a mockup. There is other design software that could also be used, but we strongly encourage you to keep it simple. Do not spend more than 15-30 minutes detailing the mock-up. Your goal should be to make a very quick and dirty diagram, to show that you have some kind of idea.

We recommend drawing each page as a rectangle, and laying out the text, buttons, and other components inside. Then, use arrows to indicate links between pages. Label each page the way you will name each function, and then put labels on the arrows to indicate what the button links. Also write down any parameters that should come into the pages.

In addition to the pages and routes, write down what fields your `State` will have, both their names and types. Think clearly about what data will need to be persistent across the entire website.

Add photo(s) or screenshot(s) of your website mock-up and labeled diagram on the next page.


Home Page
log history stats calc
log workout
distance
time
rate
type
History
workout #
"
"
total meters
avg split
best 2k
workout count
distance
time
calculate
Shows distance
time, split
& watts
save_workout
distance,
time,
rate
workout type
split + watts
PR check
update best

2) Implementing Requirements

This table should serve as a draft for how you will implement the requirements into your final project. You may change your ideas later on, but your final site must fulfill all the requirements.

Requirement	**Implementation**
Purpose/Functionality A 1-2 sentence description of the purpose or functionality of your site	**Its an erg workout log for rowers. Users can enter their workout's distance, time, rate, and type, then the site calculates their 500m split average and watt average. It keeps a history of these workouts and tracks best 2k and overall stats.**
State Dataclass Attributes Minimum of 4 attributes, one of which must be a list/dict of instances of your second dataclass. Briefly describe each attribute, its type, and how it will be meaningfully modified throughout the site.	workouts:list[Workout -> list of every logged workout. A function will be used to append Workouts into the list Best_2k : float -> the user's fastest 2k time. Updates when a logged 2k is faster Username : str -> the rower's name. Set or changed via user input Filter_type : str-> which workouts are shown on the history page. User picks which workout type to display
Second Dataclass Your second dataclass, including its attributes, their types, and how they will be meaningfully modified (if they are modified)	Workout Distance : int Time_seconds : float Stroke_rate : int Workout_type : str
Routes (minimum 7)	Index Save_name

The names of your route functions	Log_workout View_history Apply_filter Calculator show_split
Pages (minimum 5) A brief description of the pages in your site	Home -> welcomes user and has buttons to other pages Log workout -> place for user to input distance time and stroke rate + a save button History -> a table of all logged workouts Stats -> shows total meters rowed, avg split, # workouts, and best 2k time Calculator -> shows calculated split and watts
User Input (minimum 3) Description of user input components (TextBox, TextArea, CheckBox, or SelectBox) and their purposes For more information on these input types, see Drafter Documentation.	Textbox username : user inputs their name Textbox distance : user inputs distance Textbox time : user inputs time Textbox stroke_rate : user inputs rate Selectbox workout_type : user selects workout type Selectbox filter_type : user selects workout type to filter
If Statements (minimum 3) Brief description of what will be	New PR check -> if new 2k time is faster, update best time

evaluated in your if statements and how they will meaningfully change the logic of the site	Saved workouts -> checks if the input is valid ie. positive numbers History filter -> if workout is the type selected in the filter, show those workouts
Loop (minimum 1) This loop must iterate over the list/dict of dataclass instances in your State attribute. Description of this loop and why it is meaningful to the site	For my total stats page, there will be a for loop to go through the list of saved workouts to add up total distance and time to calculate averages.

LLM Conversation

If you used an LLM in any part of this work, copy and paste every part of that conversation below. Otherwise, you may be given a zero for the assignment. Obviously, these words will not all be yours, and this is the exception to the requirement at the top of this assignment that all responses be in your own words. If you did not use an LLM to assist you in this assignment, simply say so below.

I need to implement routes and pages for my website I'm creating for an erg tracker. give a brief overview of routes and pages so i can understand them for my project

A **route** is a Python function. A **page** is what that function returns and what the user sees.

Routes

A route is a function with `@route` above it. Drafter runs it when the user visits the site or clicks a button.

- Every route takes `state` as its first parameter, so it can read and change your data.
- Every route returns a `Page`.
- If the button that triggers the route is on a page with input fields, the route also takes those inputs as parameters. The parameter names must match the input names exactly.

Pages

A `Page` is built from two things:

1. **The state.** Drafter carries it forward to the next route.
2. **A list of components**, displayed top to bottom: text, `TextBox`, `SelectBox`, `Button`, `Table`, and so on.

A `Button` links to another route. When it's clicked, Drafter collects whatever the user typed into the inputs on that page and passes those values to the route.

How they could connect in an erg tracker

1. The user is on the Log Workout page, which `log_workout(state)` returned. It has `TextBox("distance")`, `TextBox("time")`, and the other inputs, plus a Save button pointing to `save_workout`.
2. The user fills in the boxes and clicks Save.
3. Drafter calls `save_workout(state, distance, time, spm, workout_type)` with the values they typed.
4. `save_workout` adds the new `Workout` to `state.workouts` and returns a confirmation `Page`.

Two kinds of routes

Display routes, like `index`, `log_workout`, `view_history`, `view_stats`, and `calculate`. They read the State and build a page, without changing anything.

- **Action routes**, like `save_name`, `save_workout`, `apply_filter`, and `delete_workout`. They take input, change the State, then either return a new page or reuse a display route, for example `return view_history(state)`.

Key rule: inputs only reach a route through a button on the same page. To save a workout, the text boxes and the Save button need to be on the same page.

Final Submission

When you are finished, download this file as a Word Document (docx) and upload the file to the appropriate assignment on Canvas.